TrES-2b

R-1501

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-2_130726` · created 2026-08-01T14:49:52+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456499.89653 +/- 0.00023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.149 +/- 0.013
Transit depth (Rp/Rs)^2	2.22 +/- 0.39 [%]
Orbital Inclination (inc)	83.98 +/- 0.46
Airmass coefficient 1 (a1)	1.018 +/- 0.024
Airmass coefficient 2 (a2)	-0.009 +/- 0.011
Scatter in the residuals of the lightcurve fit is	2.33 %
Best Comparison Star	#2 - [390, 216]
Optimal Aperture	3.31
Optimal Annulus	5.47
Transit Duration (day)	0.0795 +/- 0.0072

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.149 ± 0.013 vs 0.1254 (deviation 1.33σ)
Duration fit vs published	0.0795 d vs 0.0746 d (deviation 0.5σ)
Mid-time O–C	-0.2 min
Depth SNR	8.4
Residual scatter	2.33 %

Light curve

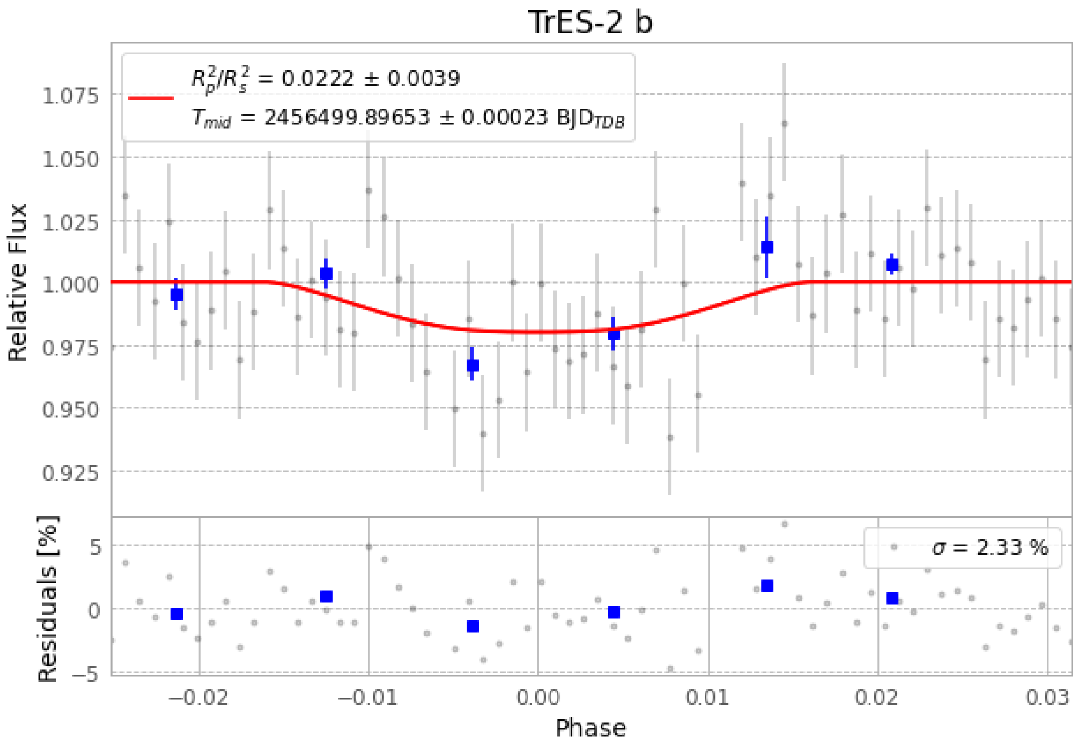

Figure 1 — FinalLightCurve_TrES-2 b_2013-07-26.png (SHA-256 667f42839d3d7cf3…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [384, 281], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 767e28d91faad0cc…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

68 FITS frames (45 MB), `TRES-2130726075713.FITS` → `TRES-2130726111812.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-130726.log (b5135ec38756…), FinalLightCurve_TrES-2 b_2013-07-26.png (667f42839d3d…), FinalLightCurve_TrES-2 b_2013-07-26.csv (68d0aae0076a…)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)